# CNN Architectures

## CS60050: Machine Learning

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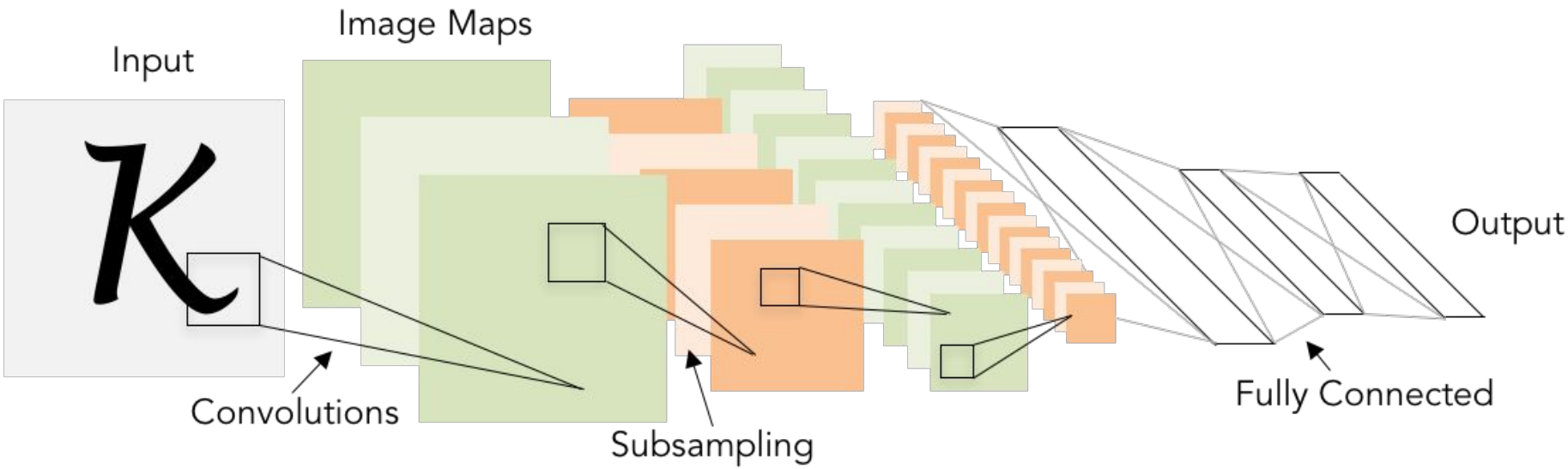
Nov 11, 2026

# Agenda

To discuss in detail about some of the highly successful deep CNN architectures

# LeNet-5

[LeCun et al., 1998]




Conv filters were 5x5, applied at stride 1
Subsampling (Pooling) layers were 2x2 applied at stride 2
i.e. architecture is [CONV-POOL-CONV-POOL-FC-FC]

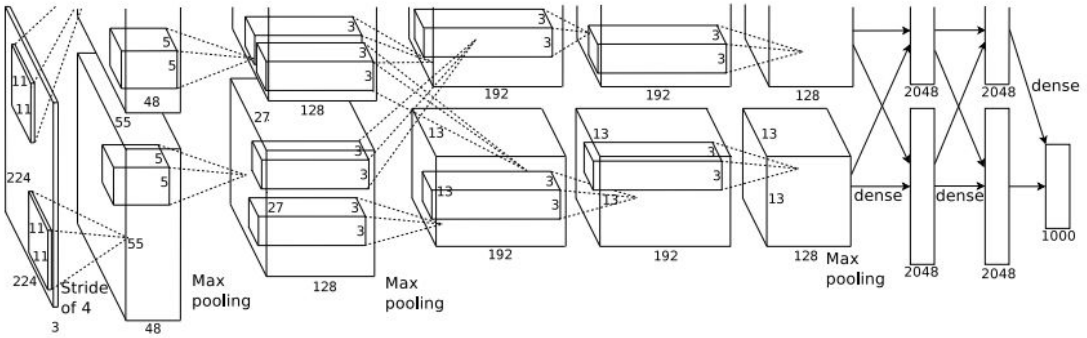
§ Citation of the paper as on Nov 10, 2026 is 91,977

Source: CS231n course, Stanford University

# AlexNet

## Case Study: AlexNet

*[Krizhevsky et al. 2012]*




**Architecture:**

CONV1
MAX POOL1
NORM1
CONV2
MAX POOL2
NORM2
CONV3
CONV4
CONV5
Max POOL3
FC6
FC7
FC8

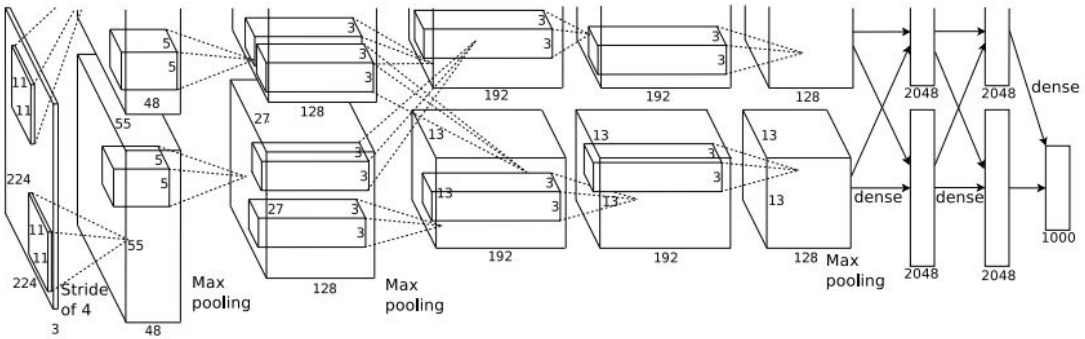
§ Citation of the paper as on Nov 10, 2026 is 1,60,669

Source: CS231n course, Stanford University

# AlexNet

## Case Study: AlexNet

*[Krizhevsky et al. 2012]*




**Architecture:**
CONV1
MAX POOL1
NORM1
CONV2
MAX POOL2
NORM2
CONV3
CONV4
CONV5
Max POOL3
FC6
FC7
FC8

Input: 227x227x3 images

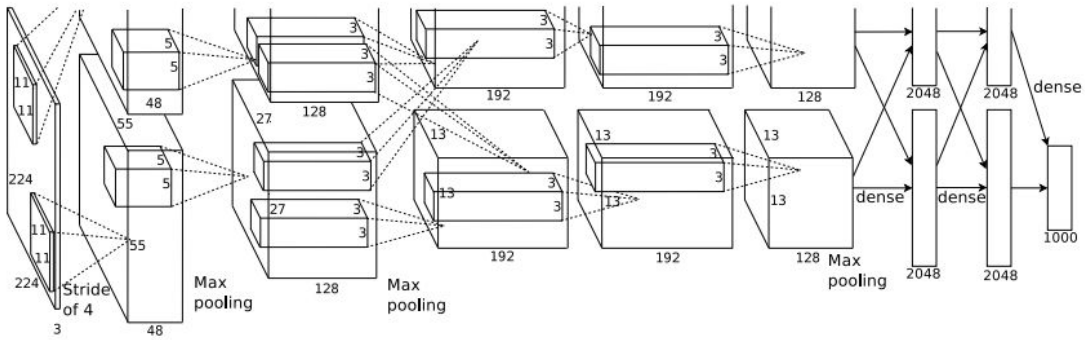
**First layer** (CONV1): 96 11x11 filters applied at stride 4
=>
Q: what is the output volume size? Hint: (227-11)/4+1 = 55

Source: CS231n course, Stanford University

# AlexNet

## Case Study: AlexNet

*[Krizhevsky et al. 2012]*




**Architecture:**
CONV1
MAX POOL1
NORM1
CONV2
MAX POOL2
NORM2
CONV3
CONV4
CONV5
Max POOL3
FC6
FC7
FC8

Input: 227x227x3 images

**First layer** (CONV1): 96 11x11 filters applied at stride 4
=>
Output volume **[55x55x96]**

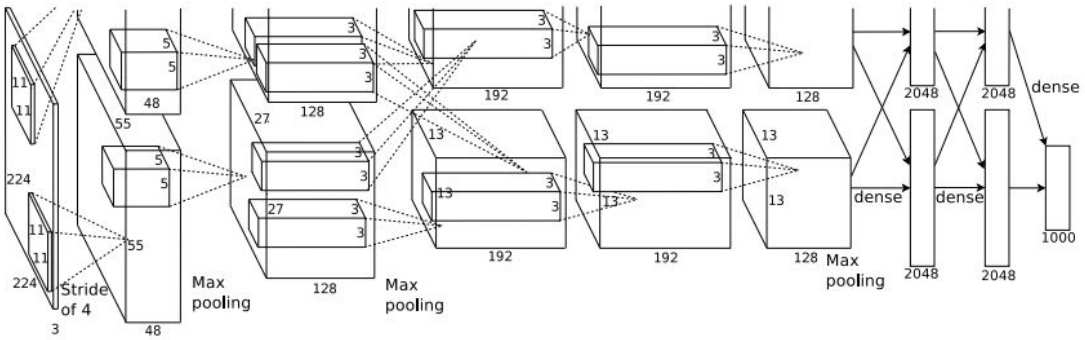
Q: What is the total number of parameters in this layer?

Source: CS231n course, Stanford University

# AlexNet

## Case Study: AlexNet

*[Krizhevsky et al. 2012]*




**Architecture:**
CONV1
MAX POOL1
NORM1
CONV2
MAX POOL2
NORM2
CONV3
CONV4
CONV5
Max POOL3
FC6
FC7
FC8

Input: 227x227x3 images

**First layer** (CONV1): 96 11x11 filters applied at stride 4
=>
Output volume **[55x55x96]**

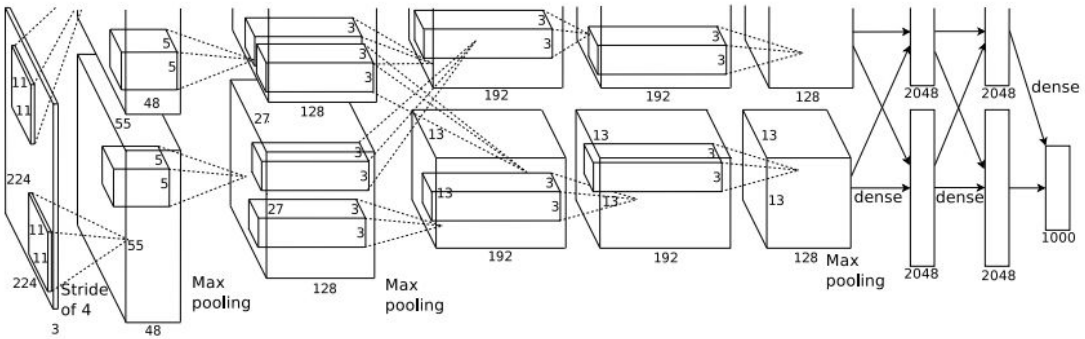
**Parameters: (11*11*3)*96 = 35K**

Source: CS231n course, Stanford University

# AlexNet

## Case Study: AlexNet

*[Krizhevsky et al. 2012]*




**Architecture:**
CONV1
MAX POOL1
NORM1
CONV2
MAX POOL2
NORM2
CONV3
CONV4
CONV5
Max POOL3
FC6
FC7
FC8

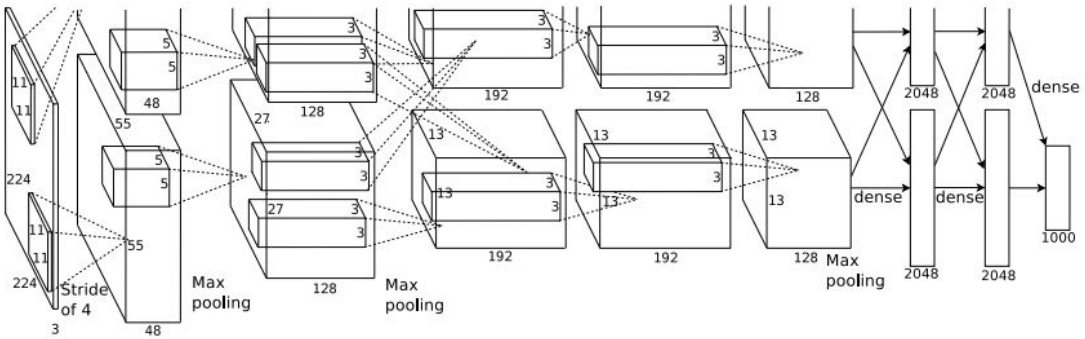
Input: 227x227x3 images
After CONV1: 55x55x96
**Second layer** (POOL1): 3x3 filters applied at stride 2
Q: what is the output volume size? Hint: (55-3)/2+1 = 27

Source: CS231n course, Stanford University

# AlexNet

## Case Study: AlexNet

*[Krizhevsky et al. 2012]*




**Architecture:**
CONV1
MAX POOL1
NORM1
CONV2
MAX POOL2
NORM2
CONV3
CONV4
CONV5
Max POOL3
FC6
FC7
FC8

Input: 227x227x3 images
After CONV1: 55x55x96
**Second layer** (POOL1): 3x3 filters applied at stride 2.
Output volume: 27x27x96

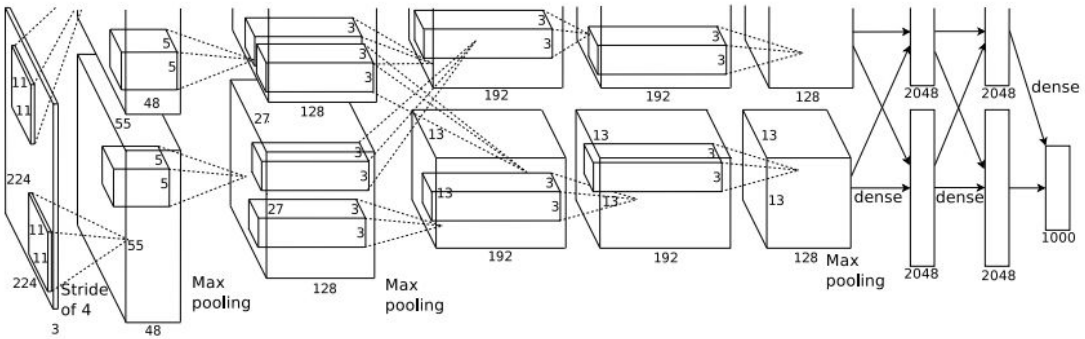
Q: what is the number of parameters in this layer?

Source: CS231n course, Stanford University

# AlexNet

## Case Study: AlexNet

*[Krizhevsky et al. 2012]*




**Architecture:**
CONV1
MAX POOL1
NORM1
CONV2
MAX POOL2
NORM2
CONV3
CONV4
CONV5
Max POOL3
FC6
FC7
FC8

Input: 227x227x3 images
After CONV1: 55x55x96
**Second layer** (POOL1): 3x3 filters applied at stride 2.
Output volume: 27x27x96

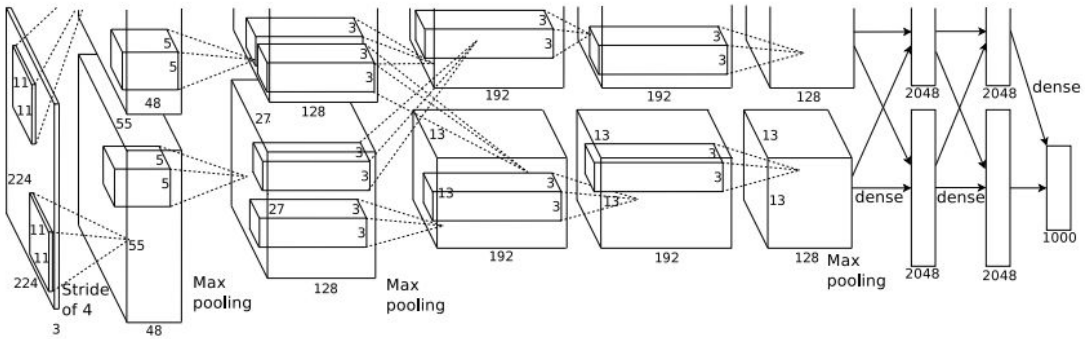
Parameters: 0!

Source: CS231n course, Stanford University

# AlexNet

## Case Study: AlexNet

*[Krizhevsky et al. 2012]*



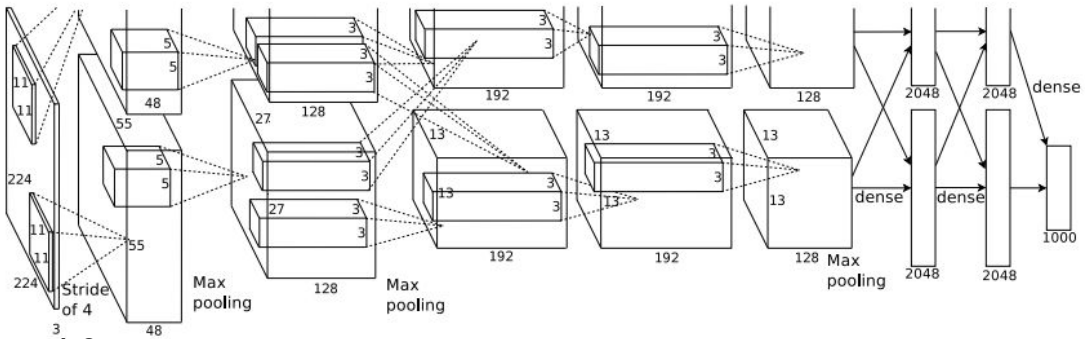

Full (simplified) AlexNet architecture:
[227x227x3] INPUT
[55x55x96] CONV1: 96 11x11 filters at stride 4, pad 0
[27x27x96] MAX POOL1: 3x3 filters at stride 2
[27x27x96] NORM1: Normalization layer
[27x27x256] CONV2: 256 5x5 filters at stride 1, pad 2
[13x13x256] MAX POOL2: 3x3 filters at stride 2
[13x13x256] NORM2: Normalization layer
[13x13x384] CONV3: 384 3x3 filters at stride 1, pad 1
[13x13x384] CONV4: 384 3x3 filters at stride 1, pad 1
[13x13x256] CONV5: 256 3x3 filters at stride 1, pad 1
[6x6x256] MAX POOL3: 3x3 filters at stride 2
[4096] FC6: 4096 neurons
[4096] FC7: 4096 neurons
[1000] FC8: 1000 neurons (class scores)

Source: CS231n course, Stanford University

# AlexNet

## Case Study: AlexNet

*[Krizhevsky et al. 2012]*

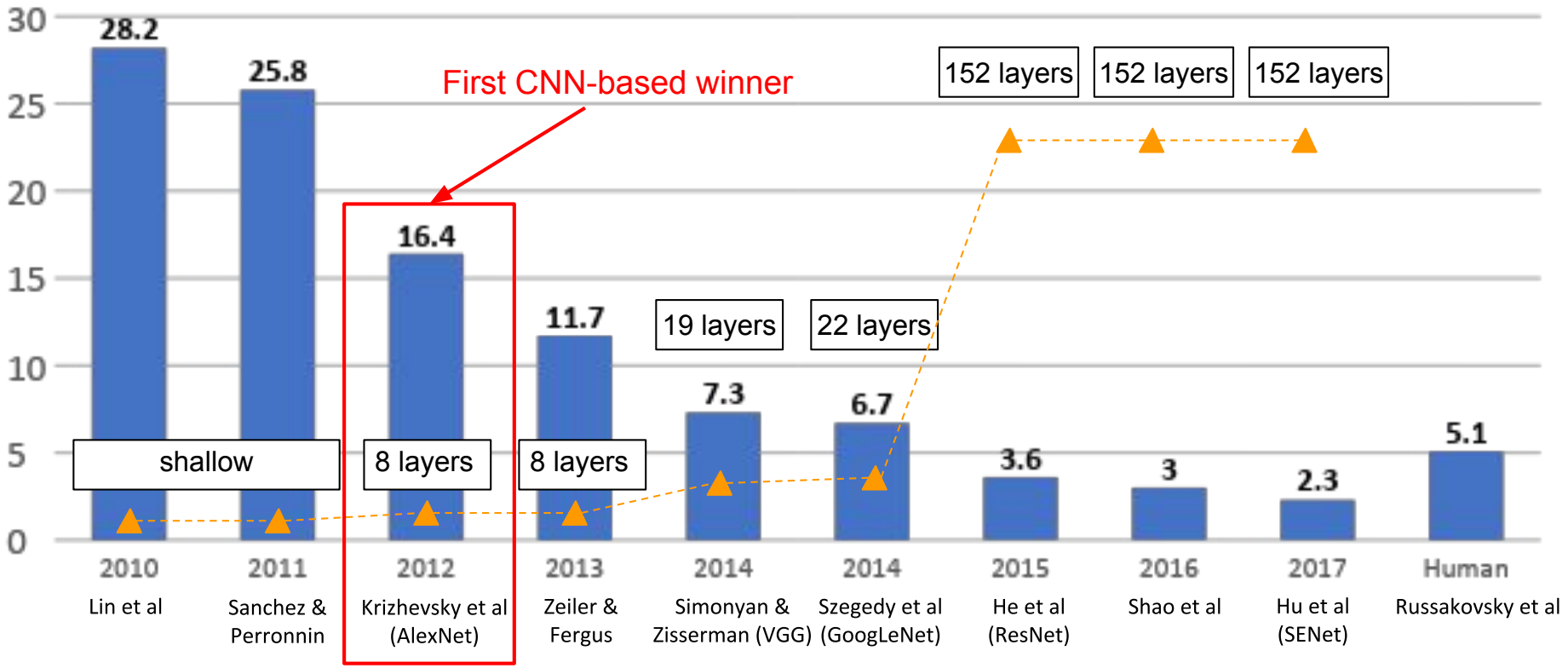



Full (simplified) AlexNet architecture:
[227x227x3] INPUT
[55x55x96] CONV1: 96 11x11 filters at stride 4, pad 0
[27x27x96] MAX POOL1: 3x3 filters at stride 2
[27x27x96] NORM1: Normalization layer
[27x27x256] CONV2: 256 5x5 filters at stride 1, pad 2
[13x13x256] MAX POOL2: 3x3 filters at stride 2
[13x13x256] NORM2: Normalization layer
[13x13x384] CONV3: 384 3x3 filters at stride 1, pad 1
[13x13x384] CONV4: 384 3x3 filters at stride 1, pad 1
[13x13x256] CONV5: 256 3x3 filters at stride 1, pad 1
[6x6x256] MAX POOL3: 3x3 filters at stride 2
[4096] FC6: 4096 neurons
[4096] FC7: 4096 neurons
[1000] FC8: 1000 neurons (class scores)

**Details/Retrospectives:**
- first use of ReLU
- used Norm layers (not common anymore)
- heavy data augmentation
- dropout 0.5
- Pooling is overlapping
- SGD Momentum 0.9
- Learning rate 1e-2, reduced by 10 manually when val accuracy plateaus
- L2 weight decay 5e-4
- 7 CNN ensemble: 18.2% -> 15.4%

Source: CS231n course, Stanford University

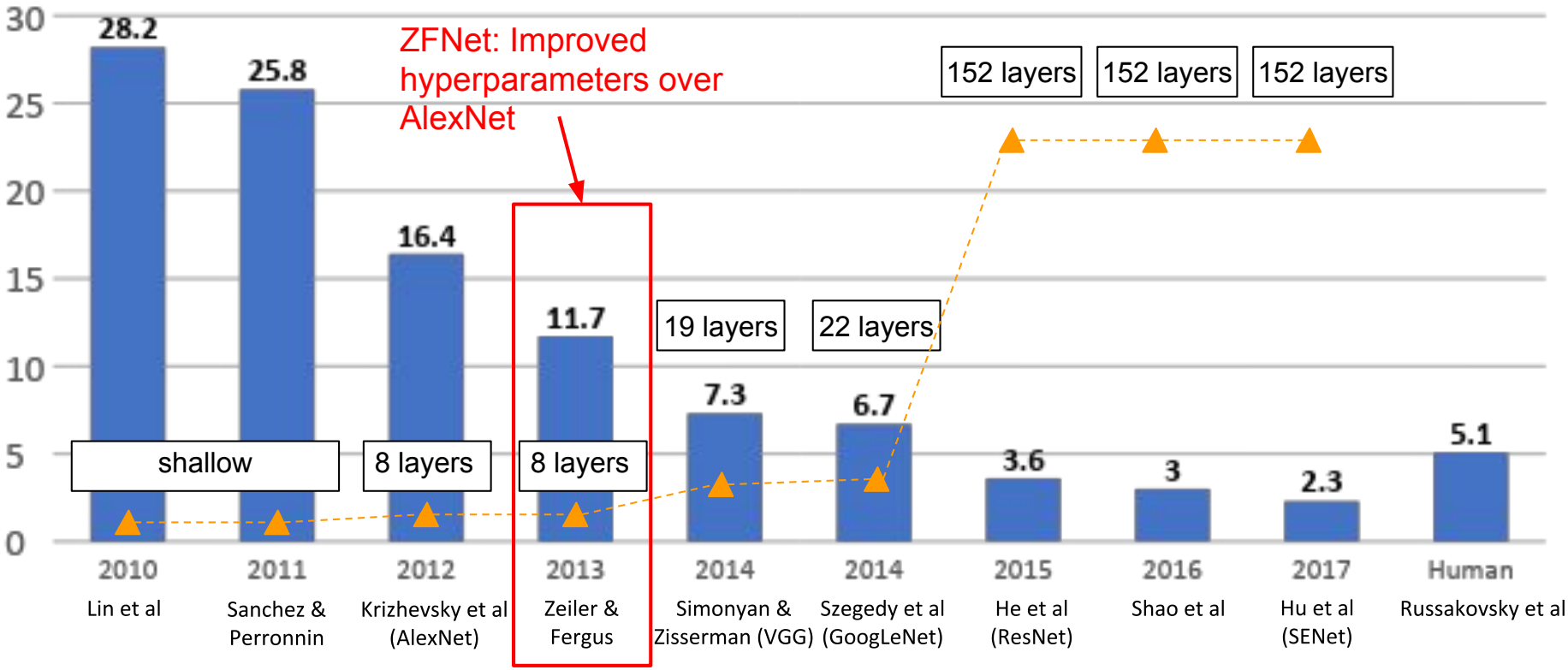
# Imagenet Leaderboard




Source: CS231n course, Stanford University

# Imagenet Leaderboard

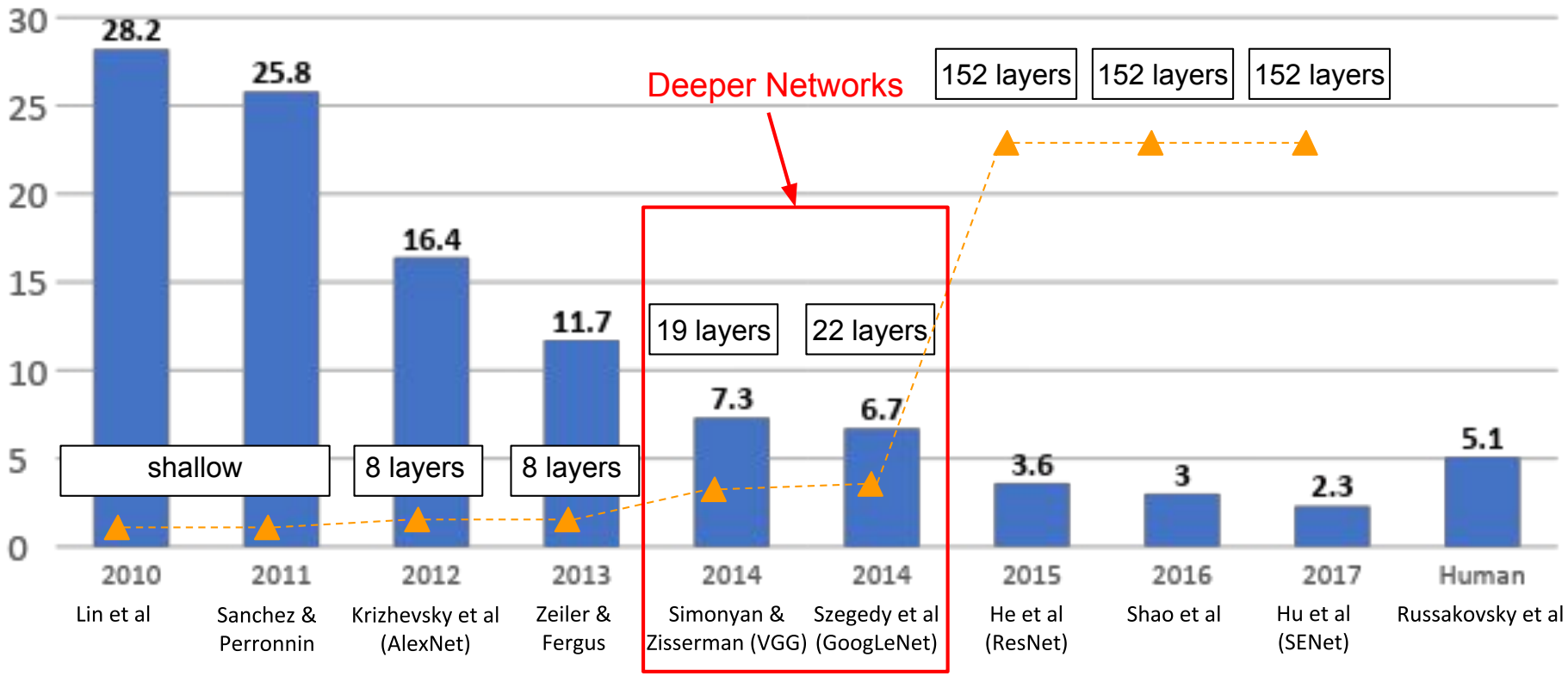
## ImageNet Large Scale Visual Recognition Challenge (ILSVRC) winners




Source: CS231n course, Stanford University

# ZFNet

## ImageNet Large Scale Visual Recognition Challenge (ILSVRC) winners




Source: CS231n course, Stanford University